9th Class • Punjab Board • PECTAA New Syllabus

Chapter 1: Physical Quantities and Measurements

Complete solved exercise — SI units, instruments, errors, significant figures and worked numericals.

Chapter 1 • Complete official exercise • 11 MCQs + 11 short + 10 constructed + 5 comprehensive + 10 numericals

What this chapter is about

- **Measurement** compares a physical quantity with an agreed standard and records a numerical value with a unit.
- **Base quantities** are independent; derived quantities combine base quantities. SI prefixes and scientific notation make very large and very small values manageable.
- **Vernier Callipers and micrometer screw gauges** improve length measurement through a vernier or circular scale. Read the main scale first, add the fractional scale reading and apply zero correction when needed.
- **Reliable measurements** require suitable instruments, correct eye position, awareness of human/random/systematic errors, and honest reporting of precision and significant figures.

Exercise coverage: All five official sections are solved in book order: A Multiple Choice, B Short Answer, C Constructed Response, D Comprehensive Questions and E Numerical Problems.

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A. Multiple Choice Questions

1. The instrument that is most suitable for measuring the thickness of a few sheets of cardboard is a:

(a) metre rule (b) measuring tape
(c) Vernier Callipers (d) micrometer screw gauge

Answer: (d) micrometer screw gauge — A micrometer screw gauge has the smallest least count and is designed for very small thicknesses.

2. One femtometre is equal to:

(a) 10^{-9} m (b) 10^{-15} m
(c) 10^{9} m (d) 10^{15} m

Answer: (b) 10^{-15} m — The prefix femto (f) means 10^{-15}, so 1 fm = 10^{-15} m.

3. A light year is a unit of:

(a) light (b) time
(c) distance (d) speed

Answer: (c) distance — A light year is the distance light travels in one year, approximately 9.46×10^{15} m.

4. Which one is a non-physical quantity?

(a) distance
(b) density
(c) colour
(d) temperature

Answer: (c) colour — Colour is a qualitative perception, not a physical quantity measured with a standard physical unit in this context.

5. When using a measuring cylinder, one precaution to take is to:

(a) check for the zero error
(b) look at the meniscus from below the level of the water surface
(c) take several readings by looking from more than one direction
(d) position the eye in line with the bottom of the meniscus

Answer: (d) position the eye in line with the bottom of the meniscus — Reading at eye level with the bottom of the water meniscus avoids parallax error.

6. Volume of water consumed by you per day is estimated in:

(a) millilitre
(b) litre
(c) kilogram
(d) cubic metre

Answer: (b) litre — Daily drinking-water volume is conveniently a few litres; a cubic metre is much too large and kilogram measures mass.

7. A displacement can is used to measure:

(a) mass of a liquid
(b) mass of a solid
(c) volume of a liquid
(d) volume of a solid

Answer: (d) volume of a solid — The volume of water displaced by a submerged solid equals the solid's volume.

8. Two rods with lengths 12.321 cm and 10.3 cm are placed side by side, the difference in their lengths is:

(a) 2.02 cm
(b) 2.0 cm
(c) 2 cm
(d) 2.021 cm

Answer: (b) 2.0 cm — 12.321 − 10.3 = 2.021 cm, but subtraction is reported to the least decimal places: one decimal place gives 2.0 cm.

9. Four students measure the diameter of a cylinder with Vernier Callipers. Which of the following readings is correct?

(a) 3.4 cm
(b) 3.475 cm
(c) 3.47 cm
(d) 3.5 cm

Answer: (c) 3.47 cm — With the usual least count of 0.01 cm, a Vernier reading is recorded to two decimal places in centimetres: 3.47 cm.

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10. Which of the following measures are likely to represent the thickness of a sheet of this book?

(a) 6×10^{-25} m (b) 1×10^{-4} m
(c) 1.2×10^{-15} m (d) 4×10^{-2} m

Answer: (b) 1×10^{-4} m — 1×10^{-4} m = 0.1 mm, a realistic paper thickness; 4×10^{-2} m = 4 cm is far too thick.

11. In a Vernier Callipers ten smallest divisions of the Vernier scale are equal to nine smallest divisions of the main scale. If the smallest division of the main scale is half millimetre, the Vernier constant is equal to:

(a) 0.5 mm (b) 0.1 mm
(c) 0.05 mm (d) 0.001 mm

Answer: (c) 0.05 mm — 1 VSD = (9/10)(0.5 mm) = 0.45 mm; Vernier constant = 1 MSD − 1 VSD = 0.50 − 0.45 = 0.05 mm.

B. Short Answer Questions

1.1. Can a non-physical quantity be measured? If yes, then how?

A non-physical quantity cannot be measured as a numerical physical quantity with a standard instrument and unit. It can only be described or compared qualitatively; an agreed rating scale or index may rank qualities such as softness or colour shade, but that does not make them physical measurements.

1.2. What is measurement? Name its two parts.

Measurement is the comparison of an unknown physical quantity with a known standard quantity of the same kind. Its two parts are a **numerical value** and a **unit**; for example, in 2.5 m, 2.5 is the numerical value and metre is the unit.

1.3. Why do we need a standard unit for measurements?

A standard unit gives every observer the same reference, so results are understandable, repeatable and comparable everywhere. It prevents confusion caused by body-based or locally varying units and allows trade, science and engineering to use consistent values.

1.4. Write the name of 3 base quantities and 3 derived quantities.

Base quantities include **length, mass and time**. Derived quantities include **area** (length × length), **speed** (length/time) and **density** (mass/volume).

1.5. Which SI unit will you use to express the height of your desk?

Use the **metre (m)**; for example, a desk may be 0.75 m high. The prefixed SI unit centimetre is also convenient, giving 75 cm.

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1.6. Write the name and symbols of all SI base units.

Base quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

1.7. Why prefix is used? Name three sub-multiple and three multiple prefixes with their symbols.

Prefixes express very small or very large values conveniently without long strings of zeros. Sub-multiples: **milli (m) = 10^{-3}, micro (μ) = 10^{-6}, nano (n) = 10^{-9}**. Multiples: **kilo (k) = 10^{3}, mega (M) = 10^{6}, giga (G) = 10^{9}**.

1.8. What is meant by: (a) 5 pm (b) 15 ns (c) 6 μm (d) 5 fs

- **5 pm:** 5 picometres = 5×10^{-12} m.
- **15 ns:** 15 nanoseconds = 15×10^{-9} s.
- **6 μm:** 6 micrometres = 6×10^{-6} m.
- **5 fs:** 5 femtoseconds = 5×10^{-15} s.

1.9. (a) For what purpose, a Vernier Callipers is used? (b) Name its two main parts. (c) How is least count found? (d) What is meant by zero error?

1. Purpose: It measures small internal/external lengths and diameters and depths with greater precision than a metre rule.

2. Main parts: The fixed **main scale** and the sliding **Vernier scale**; jaws and a depth rod are provided for different measurements.

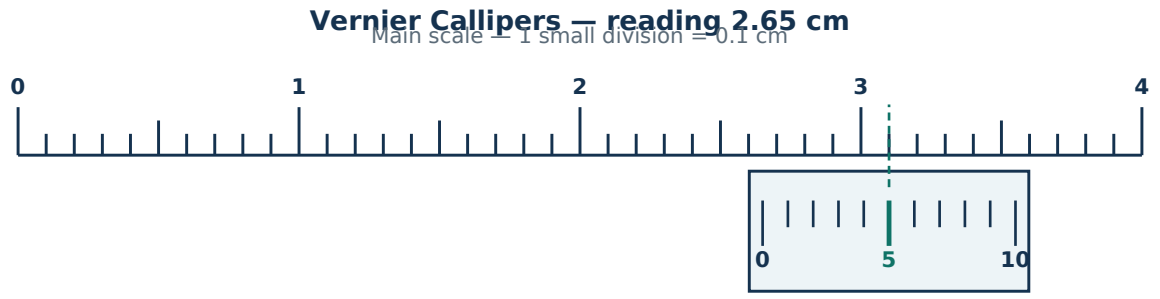
3. Least count: LC = value of 1 main-scale division − value of 1 Vernier division; equivalently, smallest main-scale division ÷ number of Vernier divisions for the usual construction.

4. Zero error: The reading shown when the jaws are fully closed and the true reading should be zero. The measured value must be corrected for it.

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1.10. State least count and Vernier scale reading as shown in the figure and hence, find the length.

Figure 1 — Original redraw of the exercise scale



Main-scale reading = 2.60 cm • coinciding Vernier division = 5

Least count = 0.01 cm; reading = 2.60 + 5 × 0.01 = 2.65 cm

The main scale has 10 divisions in 1 cm, so 1 MSD = 0.1 cm. With 10 Vernier divisions, **least count = 0.1 cm ÷ 10 = 0.01 cm**.

Main-scale reading = 2.60 cm

Coinciding Vernier division = 5

Vernier contribution = 5 × 0.01 cm = 0.05 cm

Length = 2.60 cm + 0.05 cm = 2.65 cm

1.11. Which reading out of A, B and C shows the correct length and why?

B, 6.0 cm, is correct. The eye at B is directly above and perpendicular to the mark, so there is no parallax. Viewing obliquely from A or C makes the endpoint appear at 5.9 cm or 6.1 cm respectively.

C. Constructed Response Questions

1.1. In what unit will you express each of the following? (a) Thickness of a five-rupee coin (b) Length of a book (c) Length of football field (d) Distance between two cities (e) Mass of five-rupee coin (f) Mass of your school bag (g) Duration of your class period (h) Volume of petrol filled in the tank of a car (i) Time to boil one litre milk

Item	Suitable unit
(a) Coin thickness	millimetre (mm)
(b) Book length	centimetre (cm)
(c) Football-field length	metre (m)
(d) Distance between cities	kilometre (km)
(e) Coin mass	gram (g)
(f) School-bag mass	kilogram (kg)
(g) Class-period duration	minute (min); SI equivalent in seconds
(h) Petrol volume	litre (L)
(i) Milk-boiling time	minute (min); SI equivalent in seconds

1.2. Why might a standard system of measurement be helpful to a tailor?

A tailor can record body and garment dimensions in the same accepted units, reproduce sizes accurately, avoid fitting mistakes and communicate measurements reliably to customers and other workers.

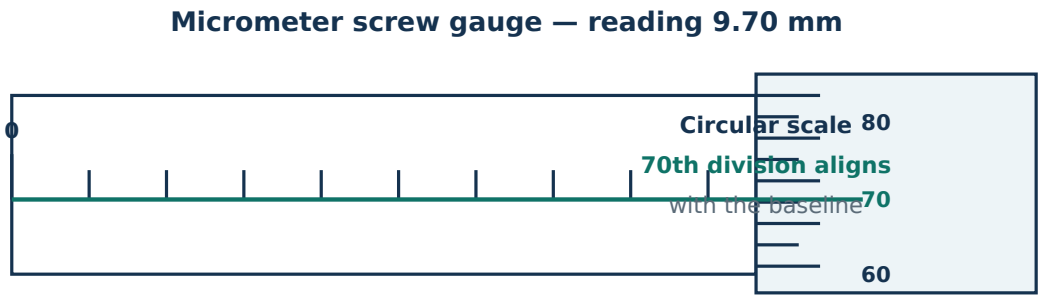
1.3. The minimum main scale reading of a micrometer screw gauge is 1 mm and there are 100 divisions on the circular scale. When thimble is rotated once, 1 mm is its measurement on the main scale. What is the least count of the instrument? The reading for thickness of a steel rod as shown in the figure. What is the thickness of the rod?

Pitch = 1 mm per rotation

Least count = pitch ÷ circular divisions = 1 mm ÷ 100 = 0.01 mm

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Figure 2 — Original redraw of the micrometer reading



Least count = pitch ÷ 100 = 1 mm ÷ 100 = 0.01 mm

Thickness = 9 mm + 70 × 0.01 mm = 9.70 mm

Main-scale reading = 9 mm

Circular-scale reading = 70 × 0.01 mm = 0.70 mm

Thickness = 9 mm + 0.70 mm = 9.70 mm

1.4. You are provided a metre scale and a bundle of pencils; how can the diameter of a pencil be measured using the metre scale with the same precision as that of Vernier Callipers? Describe briefly.

Place 10 identical pencils tightly side by side without gaps and measure the total width with the metre scale. Divide by 10. A 1 mm uncertainty in the total width then becomes 0.1 mm per pencil, equal to 0.01 cm, the usual Vernier least count. More generally, use N pencils so that (metre-rule least count)/N equals the desired least count.

1.5. The end of a metre scale is worn out. Where will you place a pencil to find the length?

Do not use the worn zero end. Place one end of the pencil at a clear graduation such as the 1 cm mark, read the graduation at the other end and subtract 1 cm from that reading.

1.6. Why is it better to place the object close to the metre scale?

Keeping the object against or close to the scale makes its endpoints align accurately with the graduations and reduces parallax and placement error.

1.7. Why a standard unit is needed to measure a quantity correctly?

It provides an unchanging, universally accepted reference. Different observers can then obtain and compare the same numerical value instead of relying on variable local or body-based units.

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1.8. Suggest some natural phenomena that could serve as a reasonably accurate time standard.

Periodic natural events can define time: the Earth's rotation for a day, the Earth's revolution around the Sun for a year, and regular atomic vibrations for extremely precise modern time standards. The event must repeat uniformly and be observable reliably.

1.9. It is difficult to locate the meniscus in a wider vessel. Why?

In a wide vessel the surface is almost flat over most of its width and capillary curvature is mainly near the walls, so a clear bottom of the meniscus is difficult to identify. A narrow measuring cylinder gives a distinct curved meniscus and allows a more precise eye-level reading.

1.10. Which instrument can be used to measure: (i) internal diameter of a test tube (ii) depth of a beaker?

Use **Vernier Callipers** for both: the **internal jaws** measure the test tube's internal diameter, and the **depth rod** measures the beaker's depth.

D. Comprehensive Questions

1.1. What is meant by base and derived quantities? Give the names and symbols of SI base units.

Base quantities are independent physical quantities chosen as the foundation of a measurement system. **Derived quantities** are obtained by multiplying or dividing base quantities according to a physical relationship.

Base quantity	Unit (symbol)	Base quantity	Unit (symbol)
Length	metre (m)	Electric current	ampere (A)
Mass	kilogram (kg)	Thermodynamic temperature	kelvin (K)
Time	second (s)	Amount of substance	mole (mol)
Luminous intensity	candela (cd)	—	—

For example, speed is derived from length and time, so its SI unit is m s^{-1}.

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1.2. Give three examples of derived unit in SI. How are they derived from base units? Describe briefly.

Derived quantity	Relationship	Derived SI unit
Speed	distance ÷ time	$m\ s^{-1}$
Density	mass ÷ volume; volume = $length^3$	$kg\ m^{-3}$
Force	mass × acceleration; acceleration = velocity change ÷ time	$kg\ m\ s^{-2}$, called newton (N)

The unit follows directly from the defining equation. Other examples include area (m^2), work (N m = joule) and power (J s^{-1} = watt).

1.3. State the similarities and differences between Vernier Callipers and micrometer screw gauge.

Similarities: Both measure small dimensions precisely, have a main and a fractional scale, require a main-scale plus fractional-scale reading, and may require zero correction.

Feature	Vernier Callipers	Micrometer screw gauge
Fractional scale	Sliding Vernier scale	Rotating circular/thimble scale
Typical least count	0.01 cm = 0.1 mm (construction may vary)	0.01 mm for 1 mm pitch and 100 divisions
Main uses	External/internal dimensions and depth	Small external thickness or diameter
Working principle	Coincidence of main and Vernier graduations	Controlled screw pitch converted into circular divisions
Typical range	Larger than a micrometer	Smaller and more specialised

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1.4. Identify and explain the reasons for human errors, random errors and systematic errors in experiments.

Error type	Reasons and behaviour	Control
Human error	Misreading a scale, wrong eye position, late/early timing, incorrect recording or calculation.	Train the observer, use correct technique, repeat readings and check records.
Random error	Unpredictable fluctuations in judgement, timing or conditions cause readings to scatter above and below the true value.	Take many readings and use their mean; improve control of conditions.
Systematic error	A consistent fault such as zero error, poor calibration or a warped scale shifts readings in one direction.	Calibrate the instrument, determine and apply correction, compare with a standard.

Repeating measurements reduces random uncertainty but does not remove a systematic zero or calibration error.

1.5. Differentiate between precision and accuracy of a measurement with examples.

Precision	Accuracy
Closeness of repeated measurements to one another.	Closeness of a measurement or mean result to the true/accepted value.
Describes reproducibility and resolution.	Describes correctness.
Example: 2.451, 2.452 and 2.451 cm are precise even if a zero error shifts all of them.	Example: a result of 2.50 cm is accurate when the accepted length is 2.50 cm.

A target analogy helps: tightly clustered shots show precision; shots clustered around the bullseye show accuracy. Good experiments aim for both.

E. Numerical Problems

1.1 Calculate the number of seconds in a (a) day (b) week (c) month and state your answers using SI prefixes.

Given: 1 day = 24 h; 1 h = 3600 s; take one month = 30 days.

$$1 \text{ day} = 24 \times 3600 = 86{,}400 \text{ s} = 86.4 \text{ ks}$$

$$1 \text{ week} = 7 \times 86{,}400 = 604{,}800 \text{ s} = 604.8 \text{ ks}$$

$$1 \text{ month} = 30 \times 86{,}400 = 2{,}592{,}000 \text{ s} = 2.592 \text{ Ms}$$

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Answer: (a) 86.4 ks; (b) 604.8 ks; (c) 2.592 Ms.

1.2 State the answers of problem 1.1 in scientific notation.

Given: Use the unrounded second values from problem 1.1.

$$86{,}400\ \text{s} = 8.64 \times 10^{4}\ \text{s}$$

$$604{,}800\ \text{s} = 6.048 \times 10^{5}\ \text{s}$$

$$2{,}592{,}000\ \text{s} = 2.592 \times 10^{6}\ \text{s}$$

Answer: (a) 8.64×10^{4} s; (b) 6.048×10^{5} s; (c) 2.592×10^{6} s.

1.3 Solve the following addition or subtraction. State your answers in scientific notation. (a) 4×10^{-4} kg + 3×10^{-5} kg (b) 5.4×10^{-6} m − 3.2×10^{-5} m

Given: Write both terms with the same power of ten before combining them.

$$(a)\ 3 \times 10^{-5} = 0.3 \times 10^{-4}$$

$$4.0 \times 10^{-4} + 0.3 \times 10^{-4} = 4.3 \times 10^{-4}\ \text{kg}$$

$$(b)\ 5.4 \times 10^{-6} = 0.54 \times 10^{-5}$$

$$0.54 \times 10^{-5} - 3.2 \times 10^{-5} = -2.66 \times 10^{-5}\ \text{m}$$

Answer: (a) 4.3×10^{-4} kg; (b) -2.66×10^{-5} m.

1.4 Solve the following multiplication or division. State your answers in scientific notation. (a) $(5 \times 10^{4}\ \text{m}) \times (3 \times 10^{-2}\ \text{m})$ (b) $(6 \times 10^{8}\ \text{kg}) \div (3 \times 10^{4}\ \text{m}^3)$

Given: Multiply/divide numerical coefficients and apply exponent rules; include the units.

$$(a)\ (5 \times 3) \times 10^{4+(-2)}\ \text{m}^2 = 15 \times 10^{2}\ \text{m}^2 = 1.5 \times 10^{3}\ \text{m}^2$$

$$(b)\ (6 \div 3) \times 10^{8-4}\ \text{kg m}^{-3} = 2.0 \times 10^{4}\ \text{kg m}^{-3}$$

Answer: (a) 1.5×10^{3} m²; (b) 2.0×10^{4} kg m⁻³.

1.5 Calculate $[(3 \times 10^{2}\ \text{kg}) \times (4.0\ \text{km})] \div (5 \times 10^{2}\ \text{s}^2)$ and state your answer in scientific notation.

Given: 4.0 km = 4.0×10^{3} m.

$$[(3 \times 10^2) \times (4.0 \times 10^3)] \div (5 \times 10^2)\ \text{kg m s}^{-2}$$

$$= (12 \times 10^5) \div (5 \times 10^2)\ \text{kg m s}^{-2}$$

$$= 2.4 \times 10^3\ \text{kg m s}^{-2}$$

Answer: 2.4×10^3 kg m s^{-2} (or N).

1.6 State the number of significant digits in each measurement: (a) 0.0045 m (b) 2.047 m (c) 3.40 m (d) 3.420×10^4 m

Given: Leading zeros are not significant; captive zeros and trailing zeros after a decimal point are significant.

(a) 0.0045 → digits 4 and 5 → 2 significant digits

(b) 2.047 → 2, 0, 4 and 7 → 4 significant digits

(c) 3.40 → trailing zero is significant → 3 significant digits

(d) 3.420×10^4 → four significant digits in 3.420

Answer: (a) 2; (b) 4; (c) 3; (d) 4.

1.7 Write in scientific notation: (a) 0.0035 m (b) 206.4×10^2 m

Given: Scientific notation has one nonzero digit before the decimal point.

(a) $0.0035\ \text{m} = 3.5 \times 10^{-3}\ \text{m}$

(b) $206.4 \times 10^2\ \text{m} = 2.064 \times 10^2 \times 10^2\ \text{m} = 2.064 \times 10^4\ \text{m}$

Answer: (a) 3.5×10^{-3} m; (b) 2.064×10^4 m.

1.8 Write using correct prefixes: (a) 5.0×10^4 cm (b) 580×10^2 g (c) 45×10^{-4} s

Given: 1 cm = 10^{-2} m; 1 kg = 1000 g; 1 ms = 10^{-3} s.

(a) $5.0 \times 10^4\ \text{cm} = 5.0 \times 10^2\ \text{m} = 0.5\ \text{km}$

(b) $580 \times 10^2\ \text{g} = 58{,}000\ \text{g} = 58\ \text{kg}$

(c) $45 \times 10^{-4}\ \text{s} = 4.5 \times 10^{-3}\ \text{s} = 4.5\ \text{ms}$

Answer: (a) 0.5 km; (b) 58 kg; (c) 4.5 ms.

1.9 Light year is a unit of distance used in Astronomy. It is the distance covered by light in one year. Taking the speed of light as 3.0×10^8 m s^{-1}, calculate the distance.

Given: c = 3.0×10^8 m s^{-1}; t = $365 \times 24 \times 3600$ s.

$$t = 365 \times 24 \times 3600 = 31{,}536{,}000 \text{ s}$$

$$\text{Distance} = ct = (3.0 \times 10^8)(3.1536 \times 10^7)$$

$$= 9.4608 \times 10^{15} \text{ m} \approx 9.46 \times 10^{15} \text{ m}$$

Answer: 1 light year ≈ 9.46×10^{15} m.

1.10 Express the density of mercury given as 13.6 g cm^{-3} in kg m^{-3}.

Given: 1 g = 10^{-3} kg and 1 cm^3 = 10^{-6} m^3.

$$13.6 \text{ g cm}^{-3} = (13.6 \times 10^{-3} \text{ kg}) \div (10^{-6} \text{ m}^3)$$

$$= 13.6 \times 10^3 \text{ kg m}^{-3} = 1.36 \times 10^4 \text{ kg m}^{-3}$$

Answer: Density of mercury = 1.36×10^4 kg m^{-3}.

F. More Important Questions (from the text)

1. State the main rules for counting significant figures with examples.

1. Nonzero digits are always significant: 26.43 has four.

2. Zeros between significant digits are significant: 20.07 has four.

3. Leading zeros before the first nonzero digit are not significant: 0.0045 has two.

4. Trailing zeros after a decimal point are significant: 3.40 has three and 3.420 has four.

5. In scientific notation, count the significant digits in the coefficient, not the power of ten: 3.420×10^4 has four.

6. For addition/subtraction, report to the least decimal places; for multiplication/division, normally report to the least number of significant figures in the data.

2. Explain positive and negative zero error and zero correction.

Close the measuring jaws or spindle gently. If the instrument does not read zero, it has a zero error.

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Zero error	Scale indication	Correction to observed reading
Positive	Fractional-scale zero is ahead of the main-scale zero; the instrument reads too much.	Subtract the positive zero error.
Negative	Fractional-scale zero is behind the main-scale zero; the instrument reads too little.	Add the magnitude of the negative zero error.

In every case: **corrected reading = observed reading − signed zero error**. Repeat the zero check before and after a measurement when high reliability is needed.

3. How can the volume of an irregular solid be found with a displacement can?

1. Fill the displacement can until water just reaches the spout and let excess water drain.

2. Place a dry measuring cylinder under the spout.

3. Lower the insoluble solid gently until it is completely submerged, without splashing; use a thread if appropriate.

4. Collect the displaced water until dripping stops and read its volume at eye level using the bottom of the meniscus.

5. The solid's volume equals the collected water volume: V = final collected volume. Convert mL to cm³ using 1 mL = 1 cm³.

The method assumes that the solid does not dissolve, react, float unless fully submerged by a suitable sinker, or trap air bubbles.

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Points to Remember

Measurement = numerical value + unit; standard SI units make results comparable.

Base quantities: length, mass, time, current, temperature, amount of substance and luminous intensity.

Scientific notation: one nonzero digit before the decimal; convert all quantities to consistent units before calculation.

Prefixes: femto 10^{-15}, nano 10^{-9}, micro 10^{-6}, milli 10^{-3}, kilo 10^{3} and mega 10^{6}.

Vernier: reading = main-scale reading + coinciding division × least count; exercise figure = 2.65 cm.

Micrometer: LC = pitch/circular divisions; exercise reading = 9 mm + 70 × 0.01 mm = 9.70 mm.

Corrected reading = observed reading − signed zero error.

Meniscus: read water at eye level with its bottom; avoid parallax.

Random errors scatter readings; **systematic errors** shift them consistently; repetition does not remove zero error.

Precision is repeatability; **accuracy** is closeness to the accepted value.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	b	3	c	4	c	5	d
6	b	7	d	8	b	9	c	10	b

How to attempt this chapter in the paper

• Always write the formula, substitute values with units and show the conversion before the final numerical answer.
• For instrument questions, write main-scale reading, fractional-scale reading, least count and zero correction separately.
• For significant figures, do not confuse leading zeros with trailing zeros after a decimal point.
• In scientific notation, make powers of ten equal before addition or subtraction and include the resulting unit.
• For practical questions, include eye position, repeated readings, calibration and a suitable precaution where relevant.

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